Molecular Basis of Inheritance

वंशगति का आण्विक आधार • सम्पूर्ण उच्च-अंक ट्यूटोरियल

Class XII NCERT Chapter 5 Syllabus Coverage: DNA Structure, Packaging, Replication, Transcription, Genetic Code, Translation, Gene Regulation (Lac Operon), Human Genome Project, and DNA Fingerprinting.

---

1. Learning Objectives and Examination Scope

1. अधिगम उद्देश्य एवं परीक्षा दृष्टिकोण

This chapter carries an extraordinarily high weightage in the NEET UG examination (consistently yielding 6 to 9 direct questions or 24 to 36 marks). The core objective is to master the chemical nature, mechanism of information storage, faithful transmission, decoding, and regulation of genetic material across prokaryotic and eukaryotic systems.

यह अध्याय राष्ट्रीय पात्रता सह प्रवेश परीक्षा (NEET UG) के दृष्टिकोण से अत्यंत महत्वपूर्ण है (प्रतिवर्ष 6 से 9 प्रश्न पूछे जाते हैं)। इसका मुख्य उद्देश्य आनुवंशिक पदार्थ की रासायनिक संरचना, सूचना का द्विगुणन, अनुलेखन, आनुवंशिक कूट, रूपांतरण तथा प्रचालेक नियमन, मानव जीनोम परियोजना और डीएनए अंगुलिछापी की विस्तृत वैज्ञानिक समझ विकसित करना है।

Key NEET Focus Dimensions:

- Quantitative base composition and Chargaff's rules; nucleotide geometry and phosphodiester linkages.
- Packaging stoichiometry: Nucleosome core particles, histone octamer charges, and higher-order chromatin folding.
- Historical experimental proofs: Griffith's transformation, Avery-MacLeod-McCarty enzyme digests, Hershey-Chase isotopic differentiation ($^{35}\mathrm{S}$ vs $^{32}\mathrm{P}$), and Meselson-Stahl density gradient centrifugation ($^{15}\mathrm{N}$ vs $^{14}\mathrm{N}$).
- Directionality of polymerases: strictly $5' \rightarrow 3'$ elongation; identification of template vs. coding strands.
- Lac operon kinetics, inducer interactions, and mutation consequences in structural vs. regulatory genes.
- Methodological contrasts: ESTs vs. sequence annotation in HGP; minisatellites (VNTRs) in DNA profiling.

2. The DNA • Deoxyribonucleic Acid

2. डीएनए • डीऑक्सीराइबोन्यूक्लिक अम्ल

DNA is a long polymer of deoxyribonucleotides. The length of a DNA molecule is defined by the number of nucleotides (or base pairs, bp) present within it, which is characteristic of every organism:

डीएनए डीऑक्सीराइबोन्यूक्लियोटाइड्स का एक लंबा बहुलक है। किसी जीव में डीएनए की लंबाई उसमें उपस्थित न्यूक्लियोटाइड्स (या क्षार युग्मों, bp) की संख्या द्वारा निर्धारित होती है, जो उस जीव का विशिष्ट लक्षण है:

Organism / Biological Entity	Genetic Material Structure	Nucleotide / Base Pair (bp) Count
Bacteriophage $\phi \times 174$	Single-stranded circular DNA	5,386 nucleotides

Organism / Biological Entity	Genetic Material Structure	Nucleotide / Base Pair (bp) Count
Bacteriophage Lambda (λ)	Double-stranded linear DNA	48,502 base pairs (bp)
Escherichia coli	Double-stranded circular DNA	4.6×10^6 bp (Length $= 1.36$ mm)
Human cell (Haploid genome, n)	Double-stranded linear DNA	3.3×10^9 bp
Human cell (Diploid somatic, $2n$)	Double-stranded linear DNA	6.6×10^9 bp (Length ≈ 2.2 metres)

2.1 Structure of a Polynucleotide Chain

2.1 पॉलीन्यूक्लियोटाइड शृंखला की संरचना

A monomeric nucleotide is composed of three distinct chemical parts:

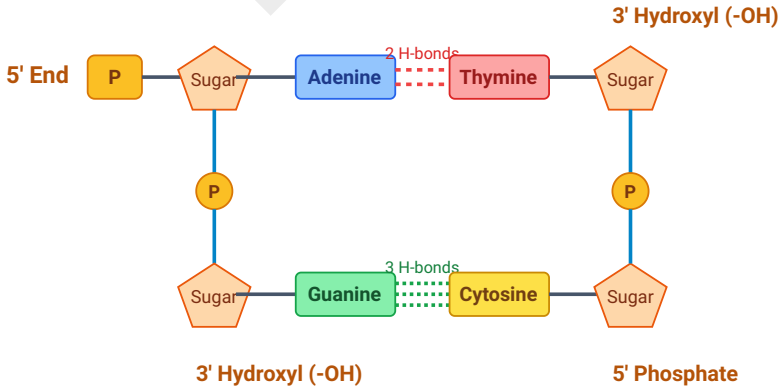
1. **Nitrogenous Base:** Heterocyclic aromatic rings classified into:
   - *Purines (Bicyclic):* Adenine (A) and Guanine (G).
   - *Pyrimidines (Monocyclic):* Cytosine (C), Uracil (U in RNA), and Thymine (5-methyl uracil in DNA).
2. **Pentose Sugar:** Deoxyribose ($C_5H_{10}O_4$) in DNA, lacking an oxygen at the $2'$ position; Ribose ($C_5H_{10}O_5$) in RNA, possessing a reactive $-OH$ group at the $2'$ position.
3. **Phosphate Group:** Phosphoric acid moiety attached to the $5'$-hydroxyl group.

> प्रत्येक न्यूक्लियोटाइड तीन घटकों से निर्मित होता है: एक नाइट्रोजनी क्षार (प्यूरीन: एडेनिन, ग्वानिन; पिरिमिडीन: साइटोसिन, थायमिन/यूरेसिल), एक पेंटोस शर्करा (डीएनए में डीऑक्सीराइबोज तथा आरएनए में राइबोज), एवं एक फॉस्फेट समूह।

Chemical Linkages of the Backbone:

- **N-glycosidic Linkage:** Formed between the nitrogenous base and the $1'$-carbon of the pentose sugar via removal of a water molecule, yielding a *nucleoside* (e.g., Deoxyadenosine, Deoxyguanosine, Deoxycytidine, Deoxythymidine).
- **Phosphoester Bond:** Formed when a phosphate group is esterified to the $5'$-OH of the nucleoside sugar to produce a *nucleotide*.
- **$3'$–$5'$ Phosphodiester Linkage:** Joins two successive nucleotides between the $3'$-OH of one sugar and the $5'$-phosphate of the next, establishing a directional backbone with a free $5'$-phosphate end and a free $3'$-hydroxyl end.

Antiparallel Complementary Double Strands of DNA with Specific Base Pairing



Helical Dimensions (B-DNA)

- Helix diameter: 2.0 nm (20 Å)
- Pitch of helix: 3.4 nm (34 Å)
- Base pairs per turn: roughly 10 bp
- Distance between bp: 0.34 nm (3.4 Å)
- Right-handed coiling
- Base-pair stacking adds stability

Structural details of the antiparallel polynucleotide chains showing phosphodiester linkages, hydrogen bonding, and Watson-Crick B-DNA helical metrics.

BOSONANCE

2.2 Salient Features of the Watson-Crick Double Helix Model

2.2 वाटसन एवं क्रिक के द्विकुंडली मॉडल के प्रमुख लक्षण

Proposed in 1953 by James Watson and Francis Crick, based on Rosalind Franklin and Maurice Wilkins' X-ray diffraction data and Erwin Chargaff's rules:

- **Two Polynucleotide Chains:** Formed with a sugar-phosphate backbone on the outside and nitrogenous bases projecting inwards.
- **Antiparallel Polarity:** One chain runs $5' \rightarrow 3'$, and the other runs in the inverted direction $3' \rightarrow 5'$.
- **Complementary Base Pairing:** Adenine always pairs with Thymine via two hydrogen bonds ($\mathrm{A} = \mathrm{T}$); Guanine pairs with Cytosine via three hydrogen bonds ($\mathrm{G} \equiv \mathrm{C}$). Consequently, a purine always faces a pyrimidine, maintaining a strictly uniform diameter of $2.0\ \mathrm{nm}$ throughout the helix.
- **Helical Metrics:** Right-handed coiling; pitch is $3.4\ \mathrm{nm}$ with $\approx 10\ \mathrm{bp}$ per turn, resulting in a distance of $0.34\ \mathrm{nm}$ ($3.4 \times 10^{-10}\ \mathrm{m}$) between adjacent base pairs.
- **Dual Stability:** Maintained by (1) complementary hydrogen bonding, and (2) stacking of hydrophobic aromatic planar base pairs over one another.

Chargaff's Equivalence Rule (Applicable ONLY to Double-Stranded DNA):

$$\frac{[\mathrm{A}]}{[\mathrm{T}]} = 1 \quad \text{and} \quad \frac{[\mathrm{G}]}{[\mathrm{C}]} = 1 \implies [\mathrm{A}] + [\mathrm{G}] = [\mathrm{T}] + [\mathrm{C}] \quad (\text{Purines} = \text{Pyrimidines})$$

NEET Warning: The base ratio $\frac{[\mathrm{A}]+[\mathrm{T}]}{[\mathrm{G}]+[\mathrm{C}]}$ is **not** necessarily 1; it is species-specific (variable). Furthermore, Chargaff's rule fails for single-stranded DNA ($\phi \times 174$) and all RNA molecules!

2.3 Francis Crick's Central Dogma

2.3 फ्रांसिस क्रिक का सेंट्रल डोग्मा (मूल सिद्धांत)

States that unidirectional genetic information flows from nucleic acids to functional proteins:

$$\mathrm{DNA} \xrightarrow{\text{Replication}} \mathrm{DNA} \xrightarrow{\text{Transcription}} \mathrm{mRNA} \xrightarrow{\text{Translation}} \text{Protein}$$

Exception: In retroviruses (e.g., Tobacco Mosaic Virus, HIV), reverse flow occurs via reverse transcription (Teminism): $\mathrm{RNA} \rightarrow \mathrm{DNA}$.

2.4 Packaging of DNA Helix in Cells

2.4 डीएनए कुंडली का संवेष्टन

The total length of double-helical DNA in a human diploid somatic cell ($6.6 \times 10^9\ \mathrm{bp}$) is calculated as:

$$\text{Length} = 6.6 \times 10^9\ \mathrm{bp} \times 0.34 \times 10^{-9}\ \mathrm{m/bp} \approx 2.24\ \text{metres}$$

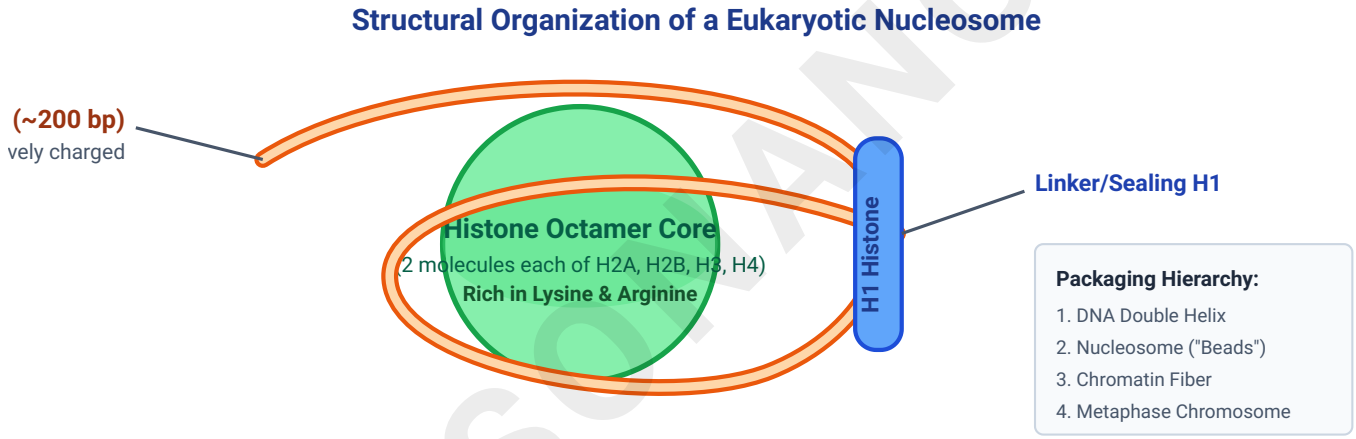
Packaging a $2.2\ \mathrm{m}$ macromolecule into a microscopic nucleus of dimension $\approx 10^{-6}\ \mathrm{m}$ represents an extraordinary condensing feat achieved differently across domains:

BOSONANCE

Prokaryotic Packaging (e.g., *E. coli*):
Lacking a defined nuclear membrane, negatively charged DNA is organized with positively charged non-histone polyamines into large looped domains anchored by proteins within a clear cytoplasmic sector termed the **nucleoid**.

Eukaryotic Packaging — The Nucleosome:
Histones are basic proteins enriched in positively charged basic amino acid residues: **Lysine** and **Arginine**.

- Eight histone molecules (2 each of $\mathrm{H2A}, \mathrm{H2B}, \mathrm{H3}, \mathrm{H4}$) assemble into a cylindrical **histone octamer** core.
- Negatively charged DNA wraps roughly 1.75 turns ($\approx 200\ \mathrm{bp}$ of DNA helix) around the octamer to form the fundamental repeating unit: the **nucleosome**.
- $\mathrm{H1}$ **Histone:** Binds at the entry/exit site of DNA, sealing the nucleosome structure.
- Under electron microscopy, chromatin resembles a **'beads-on-a-string'** architecture.
- Higher-order folding beyond nucleosomes requires non-histone chromosomal (NHC) proteins to yield condensed metaphase chromosomes.



Organization of a nucleosome showing the histone octamer core, wrapping DNA helix, and sealing H1 histone protein.

Feature	Euchromatin	Heterochromatin
Packaging Density	Loosely packed chromatin fibers	Densely, tightly packed chromatin fibers
Staining Property	Stains light under nuclear stains	Stains dark under nuclear stains
Transcriptional Status	**Transcriptionally active** (Permits RNA Pol access)	**Transcriptionally inactive / inert**

3. The Search for Genetic Material

3. आनुवंशिक पदार्थ की खोज

3.1 Griffith's Transformation Experiment (1928)

3.1 ग्रिफ़िथ का रूपांतरण प्रयोग (1928)

Frederick Griffith experimented with *Streptococcus pneumoniae* (Pneumococcus):

- **S Strain (Smooth):** Produces a smooth mucous (polysaccharide) capsule; *virulent* (causes lethal pneumonia in mice).
- **R Strain (Rough):** Non-capsulated; produces rough colonies; *non-virulent* (mice survive).

Griffith's Experimental Series:
1. Live S strain $\rightarrow$ Inject into mice $\rightarrow$ Mice Die
2. Live R strain $\rightarrow$ Inject into mice $\rightarrow$ Mice Live
3. Heat-killed S strain $\rightarrow$ Inject into mice $\rightarrow$ Mice Live
4. Heat-killed S strain + Live R strain $\rightarrow$ Inject into mice $\rightarrow$ Mice Die!

Post-mortem recovery: Living, virulent S-strain bacteria were harvested from the dead mice. Griffith deduced that a non-living "**transforming principle**" was transferred from the heat-killed S cells into live R cells, conferring upon them the enzymatic capacity to synthesize a polysaccharide capsule and become virulent. However, the biochemical nature remained unidentified.

3.2 Biochemical Characterization (Avery, MacLeod, and McCarty, 1933–44)

3.2 रूपांतरण सिद्धांत का जैव-रासायनिक लक्षण-वर्णन

Avery, MacLeod, and McCarty purified biomolecules from heat-killed S-cells to determine what converted R to S:

- Heat-killed extract treated with **Proteases** $\rightarrow$ Transformation still occurred (Protein is NOT the transforming principle).
- Heat-killed extract treated with **RNase** $\rightarrow$ Transformation still occurred (RNA is NOT the transforming principle).
- Heat-killed extract treated with **DNase** $\rightarrow$ **Transformation was completely inhibited!**

Conclusion: DNA is the substance responsible for hereditary transformation. However, many biologists remained unconvinced until 1952.

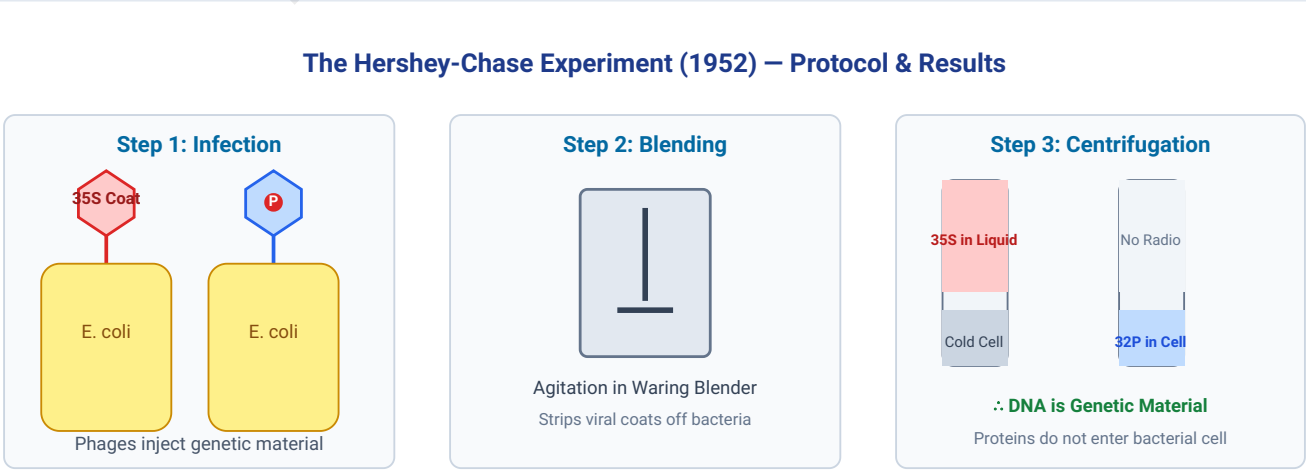
3.3 Hershey-Chase Unequivocal Proof (1952)

3.3 हर्षे एवं चेस का असंदिग्ध प्रमाण (1952)

Alfred Hershey and Martha Chase provided unequivocal proof that DNA is the genetic material using Bacteriophage $\mathrm{T2}$ infecting *E. coli*:

Batch	Radioactive Isotope Added	Biomolecule Specifically Labeled	Reason for Specificity
Batch 1	Radioactive Phosphorus ($^{32}\mathrm{P}$)	**Viral DNA**	DNA contains phosphorus, but protein capsules lack phosphorus.
Batch 2	Radioactive Sulfur ($^{35}\mathrm{S}$)	**Viral Protein Coat**	Proteins contain sulfur (in methionine/cysteine), but DNA contains zero sulfur.

BOSONANCE



Hershey-Chase protocol showing infection, blending, and centrifugation of phage-infected bacteria with selective $^{35}\mathrm{S}$ and $^{32}\mathrm{P}$ tracing.

3.4 Criteria for Genetic Material (DNA versus RNA)

3.4 आनुवंशिक पदार्थ की कसौटियां (डीएनए बनाम आरएनए)

To act as a genetic material, a molecule must fulfill four fundamental evolutionary criteria:

Criterion	DNA Property	RNA Property
1. Self-Replication	Capable via exact complementary base pairing.	Capable via complementary base pairing.
2. Chemical & Structural Stability	**Highly stable:** Double-stranded; lacks $2'$-OH; possesses thymine instead of uracil; resisted heat in Griffith's experiment.	**Labile & degradable:** $2'$-OH group present at every nucleotide makes it reactive; acts as a catalytic enzyme (ribozyme).
3. Mutational Scope for Evolution	Mutates slowly; possesses robust enzymatic repair mechanisms.	Mutates at a faster rate; RNA viruses (e.g., retroviruses) evolve rapidly.
4. Mendelian Expression	Indirect expression: dependent on RNA intermediates to synthesize proteins.	Direct expression: codes directly for polypeptide synthesis.

Conclusion: DNA is structurally superior for **stable storage** of genetic information; RNA is preferred for **transmission** of information.

4. DNA Replication

4. डीएनए प्रतिकृतियन

4.1 Semiconservative Replication Scheme

4.1 अर्ध-संरक्षी प्रतिकृतियन योजना

Watson and Crick famously deduced: *"It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material."* The two strands unwind, each acting as a template for synthesizing a brand-new complementary strand. Every daughter duplex contains one parental strand and one newly synthesized strand.

4.2 Experimental Proof — Meselson and Stahl (1958)

4.2 मेसेल्सन एवं स्टाल का प्रायोगिक प्रमाण

- Cultured *E. coli* in $^{15}NH_4Cl$ (^{15}N is a heavy stable isotope of nitrogen, **not radioactive**) for many generations until all cellular DNA contained ^{15}N.
- Transferred cells into normal $^{14}NH_4Cl$ medium; sampled DNA at definite time intervals (*E. coli* generation time $= 20$ minutes).
- Separated DNA using **cesium chloride ($CsCl$) density gradient centrifugation**:
  - *Generation 0 (0 min):* 100% Heavy DNA ($^{15}N/^{15}N$), sedimenting at the bottom.
  - *Generation 1 (20 min):* 100% Hybrid intermediate density DNA ($^{15}N/^{14}N$).
  - *Generation 2 (40 min):* 50% Hybrid ($^{15}N/^{14}N$) and 50% Light ($^{14}N/^{14}N$).
  - *Generation 3 (60 min):* 25% Hybrid and 75% Light.
  - *Generation 4 (80 min):* 12.5% Hybrid ($2/16$) and 87.5% Light ($14/16$).

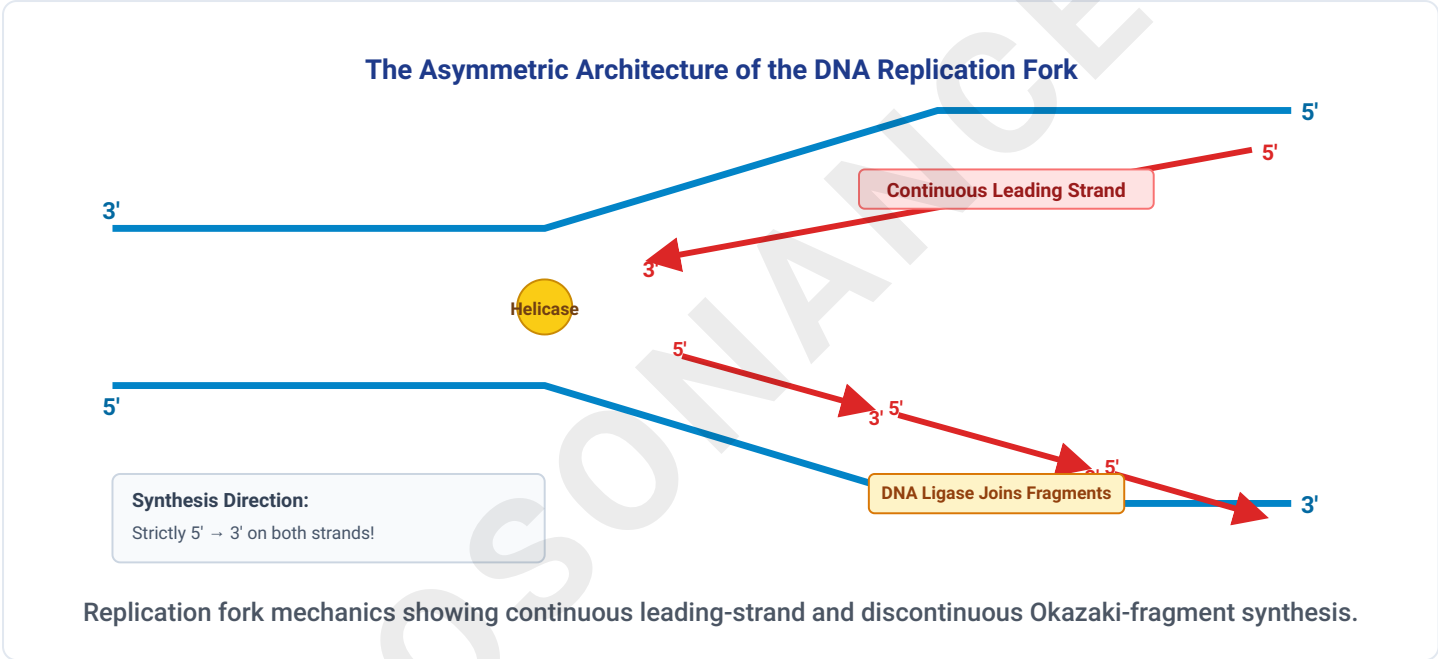
Taylor's Experiment on Higher Plants (1958):
J. Herbert Taylor and colleagues proved semiconservative replication in chromosomes of *Vicia faba* (broad bean / faba bean) using **radioactive thymidine** to track newly synthesized chromatids during root-tip cell division.

BOSONANCE

4.3 The Replication Machinery and Enzymes

4.3 प्रतिकृतियन तंत्र एवं एंजाइम

- **DNA-Dependent DNA Polymerase:** Highly efficient enzyme that synthesizes new DNA along a template. In *E. coli* (4.6×10^6 bp), replication finishes within 18 minutes, requiring an average polymerisation rate of ≈ 2000 bp/second with near-zero error!
- **Dual Role of Deoxyribonucleoside Triphosphates (dNTPs):**
  1. Act as monomeric *substrates* (dATP, dGTP, dCTP, dTTP).
  2. Supply *energy* for polymerization via high-energy pyrophosphate bonds ($\mathrm{P} \sim \mathrm{P}$), analogous to ATP.
- **Directional Constraint:** DNA polymerases strictly catalyse synthesis in the $5' \rightarrow 3'$ direction only. They cannot initiate synthesis *de novo* and require an RNA primer and a designated **Origin of Replication (ori)**.
- **Replication Fork Dynamics:** Because unwinding the whole chromosome requires prohibitive energy, replication occurs within a small localized Y-shaped unzipped window called the replication fork:
  - *Continuous / Leading Strand:* Synthesized continuously on the template with $3' \rightarrow 5'$ polarity.
  - *Discontinuous / Lagging Strand:* Synthesized discontinuously as short segments (**Okazaki fragments**) on the template with $5' \rightarrow 3'$ polarity. The fragments are covalently joined by **DNA ligase**.



Replication fork mechanics showing continuous leading-strand and discontinuous Okazaki-fragment synthesis.

5. Transcription

5. अनुलेखन

Transcription is the enzymatic process of copying genetic information from one strand of DNA into an RNA polymer. Governed by the rule of complementarity, Uracil pairs with Adenine in place of Thymine.

Why is Only One DNA Strand Transcribed?

1. If both strands were copied, they would code for two distinct RNA molecules with differing nucleotide sequences, yielding two different proteins from one DNA segment and confounding genetic regulation.
2. The two simultaneously produced complementary RNA strands would hybridize into a double-stranded RNA (dsRNA), terminating translation and rendering transcription completely futile.

BOSONANCE

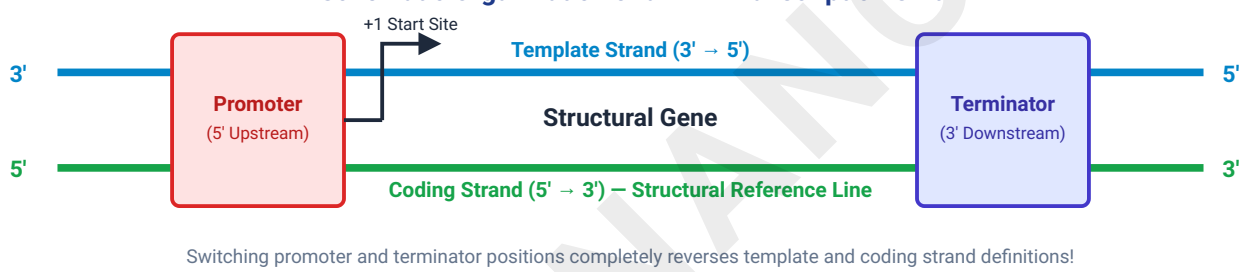
5.1 The Transcription Unit and Strand Conventions

5.1 अनुलेखन एकक एवं रज्जुक परंपरा

A transcription unit in DNA contains three structural landmarks:

- **Promoter:** Binding site for RNA polymerase located at the $5'$**-end (upstream)** relative to the coding strand.
- **Structural Gene:** The flanked region coding for functional RNA or polypeptide.
- **Terminator:** Defines the termination boundary located at the $3'$**-end (downstream)** relative to the coding strand.

Strand Name	Polarity	Role in Transcription	Sequence Comparison with Transcribed RNA
Template Strand (Non-coding / Antisense)	$3' \rightarrow 5'$	Serves as physical template for RNA polymerase.	Complementary and antiparallel to RNA.
Coding Strand (Non-template / Sense)	$5' \rightarrow 3'$	Does NOT code; displaced during elongation. *All reference landmarks (promoter upstream, terminator downstream) are named after this strand!*	**Identical** to transcribed RNA, except Thymine (T) is replaced by Uracil (U).



Transcription unit flanked by 5' upstream promoter and 3' downstream terminator relative to the coding strand.

BOSONANCE

5.2 Transcription in Prokaryotes vs. Eukaryotes

5.2 प्रोकैरियोट्स बनाम यूकैरियोट्स में अनुलेखन

Bacterial Transcription:

- A **single DNA-dependent RNA polymerase** transcribes all three major functional RNAs: mRNA (informational template), tRNA (adapter), and rRNA (structural and catalytic).
- The core RNA polymerase catalyzes only **elongation**. It transiently associates with **sigma factor (σ)** for *initiation* and with **rho factor (ρ)** for *termination*.
- Because prokaryotes lack compartmentalized nuclear membranes and mRNA requires no post-transcriptional processing, **coupled transcription-translation** commonly occurs!

Eukaryotic Complexities:

1. **Division of Labor Among Nuclear RNA Polymerases:**
   - **RNA Polymerase I:** Transcribes rRNAs ($28\mathrm{S}, 18\mathrm{S}, 5.8\mathrm{S}$).
   - **RNA Polymerase II:** Transcribes precursor of mRNA, designated as **heterogeneous nuclear RNA (hnRNA)**.
   - **RNA Polymerase III:** Transcribes tRNA, $5\mathrm{S}$ rRNA, **and** snRNAs (small nuclear RNAs).
2. **Split-Gene Architecture & Processing of hnRNA:** Primary eukaryotic transcripts possess coding expressed sequences (**exons**) interrupted by non-coding intervening sequences (**introns**). Maturation requires three post-transcriptional modifications:
   - **Splicing:** Introns are excised and exons are ligated in a defined order by spliceosomes.
   - **Capping:** Addition of an unusual nucleotide — **methyl guanosine triphosphate ($\mathrm{m^7G_{ppp}}$)** — to the $5'$-end of hnRNA.
   - **Tailing (Polyadenylation):** Addition of 200–300 adenylate residues to the $3'$-end in a template-independent manner.

6. Genetic Code, Mutations, and Translation

6. आनुवंशिक कूट, उत्परिवर्तन एवं रूपांतरण (अनुवादन)

6.1 Deciphering the Triplet Code

6.1 त्रिक कूट का कूटवाचन

Physicist **George Gamow** calculated that since only 4 nitrogenous bases exist to code for 20 primary amino acids, a single base ($4^1 = 4$) or doublet code ($4^2 = 16$) is insufficient. A **triplet code** generates $4^3 = 64$ possible codons, providing ample capacity.

- **Har Gobind Khorana:** Chemically synthesized defined homopolymers and copolymers (e.g., $\mathrm{UGUGUG}\ldots$).
- **Marshall Nirenberg:** Developed cell-free protein-synthesizing systems, deciphering that UUU codes for Phenylalanine.
- **Severo Ochoa:** Discovered polynucleotide phosphorylase, enabling template-independent enzymatic synthesis of RNA with defined base sequences.

6.2 Six Salient Features of the Genetic Code

6.2 आनुवंशिक कूट की छह प्रमुख विशेषताएं

1. **Triplet Nature:** 61 codons code for amino acids; 3 codons — $\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$ — do not code for amino acids and function as stop/nonsense codons.
2. **Degeneracy:** Most amino acids are specified by more than one codon (e.g., Leucine, Serine, and Arginine each have 6 codons).
3. **Contiguous Reading:** Codons are read on mRNA continuously from $5' \rightarrow 3'$ without commas, pauses, or punctuation.

BOSONANCE

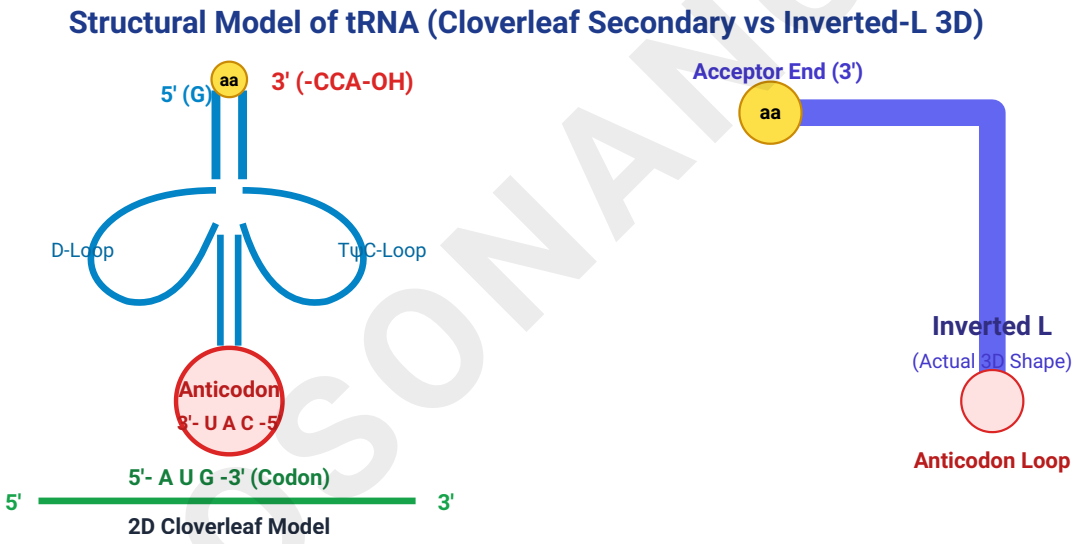
4. **Near Universality:** From bacteria to humans, UUU specifies Phenylalanine (rare exceptions exist in human mitochondrial codons and ciliate protozoans).
5. **Dual Function of AUG:** Codes for **Methionine (Met)** and serves universally as the **Initiator (Start) Codon**.
6. **Unambiguous & Specific:** One particular codon codes for one and only one amino acid.

6.3 tRNA — The Adapter Molecule

6.3 टी-आरएनए — अनुकूलक अणु

Hypothesized by Francis Crick because amino acids lack intrinsic structural capability to directly read nucleotide sequences on mRNA:

- **Anticodon Loop:** Contains three complementary bases that hydrogen-bond antiparallelly to the mRNA codon.
- **Amino Acid Acceptor Stem:** The unlooped $3'$-terminal with sequence $-\mathrm{CCA}-3'$ that covalently binds an activated amino acid.
- **Conformation:** Secondary structure resembles a **cloverleaf**; true tertiary $3\mathrm{D}$ crystal structure resembles an **inverted L**.
- **Specificity:** Distinct tRNAs exist for each amino acid, plus a unique **initiator tRNA** for start codon AUG. **No tRNAs exist for stop codons** ($\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$).



Comparison of the secondary 2D cloverleaf model of tRNA with its actual compact 3D inverted-L tertiary crystal structure.

6.4 Translation Machinery & Stepwise Kinetics

6.4 रूपांतरण तंत्र एवं चरणबद्ध कार्यप्रणाली

1. **Aminoacylation (Charging) of tRNA:** Amino acids are activated in the presence of ATP and covalently coupled to their cognate tRNA via aminoacyl-tRNA synthetase:
$$\mathrm{Amino\ Acid} + \mathrm{ATP} \rightarrow \mathrm{Aminoacyl\text{-}AMP} + \mathrm{PP_i}$$
$$\mathrm{Aminoacyl\text{-}AMP} + \mathrm{tRNA} \rightarrow \mathrm{Aminoacyl\text{-}tRNA} + \mathrm{AMP}$$
2. **Ribosome Factory:** Consists of structural rRNAs and ≈ 80 distinct proteins. In prokaryotes ($70\mathrm{S}$): $50\mathrm{S}$ large subunit and $30\mathrm{S}$ small subunit. *Crucial Ribozyme:* $23\mathrm{S}$ **rRNA** in bacteria acts as a catalytic **peptidyl transferase** enzyme forming peptide bonds!
3. **Initiation:** Small ribosomal subunit binds mRNA at $5'$-leader sequence; initiator tRNA recognizes AUG; large subunit attaches.
4. **Elongation:** Ribosome moves from codon to codon along mRNA in the $5' \rightarrow 3'$ direction, forming sequential peptide bonds between adjacent charged tRNAs.

BOSONANCE

5. **Termination:** When a stop codon ($\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$) enters the A-site, a **Release Factor** binds, hydrolyzing the finished polypeptide chain.
6. **Untranslated Regions (UTRs):** Sequences of mRNA located before the start codon (at $5'$-end) and after the stop codon (at $3'$-end) that are not translated into protein, but are essential for **efficient translation**.

7. Regulation of Gene Expression • The Lac Operon

7. जीन अभिव्यक्ति का नियमन • लैक प्रचालेक (लैक ऑपेरॉन)

In eukaryotes, gene expression can be regulated at four levels:

- Transcriptional level (formation of primary transcript).
- Processing level (regulation of splicing/capping/tailing).
- Transport level (transport of mature mRNA from nucleus to cytoplasm).
- Translational level (protein synthesis initiation and efficiency).

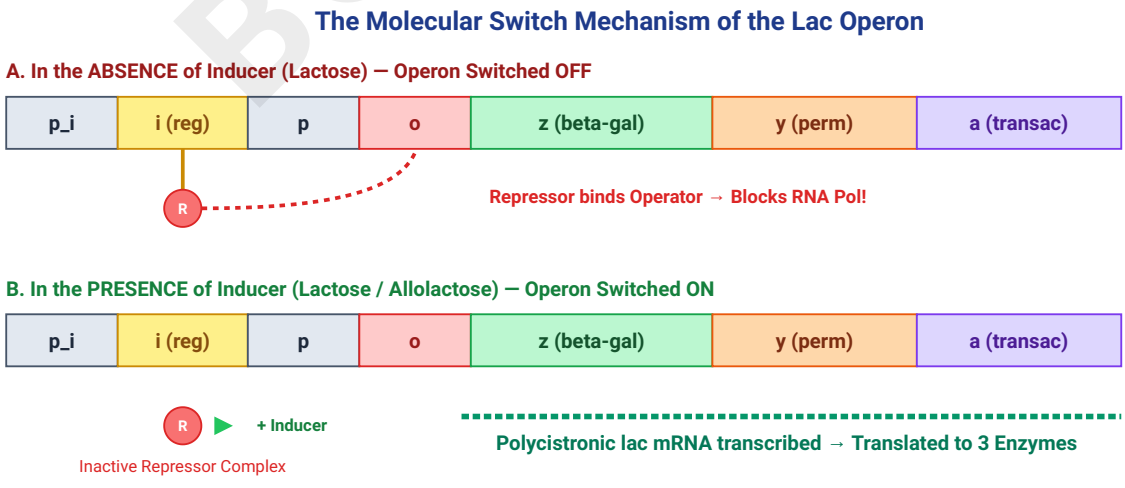
In prokaryotes, predominant control is exerted at the **initiation of transcription**.

7.1 The Lac Operon Model (Jacob and Monod)

7.1 लैक ऑपेरॉन मॉडल (जैकोब एवं मोनोड)

Elucidated by François Jacob (geneticist) and Jacques Monod (biochemist) in *E. coli* as the prototype inducible operon:

Gene / Element	Type	Encoded Product / Biological Function
i **Gene**	Regulatory gene (Constitutive)	Codes for the **Lac Repressor protein** (all-the-time synthesis; i stands for *inhibitor*, NOT inducer!).
p **Element**	Promoter	Binding site for DNA-dependent RNA polymerase.
o **Element**	Operator	Binding site for active repressor protein; controls transcription initiation.
z **Gene**	Structural gene	Codes for β**-galactosidase**; hydrolyzes lactose into **Glucose + Galactose**.
y **Gene**	Structural gene	Codes for β**-galactoside permease**; increases membrane permeability to lactose.
a **Gene**	Structural gene	Codes for β**-galactoside transacetylase**; transfers acetyl groups.



Negative inducible control of the lac operon: repressor inactivation by inducer allows transcription of structural genes z, y, and a.

BOSONANCE

NEET High-Yield Nuances of the Lac Operon:

- **Basal Level Requirement:** A very low basal level of lac operon expression must ALWAYS be present in *E. coli*; otherwise, permease would be completely absent and exogenous lactose could never enter the cell!
- **Physiological Inducer:** *Allolactose* (an isomer generated by β-galactosidase) is the genuine physiological inducer.
- **Substrate Specificity:** Glucose or galactose *cannot* act as inducers. When glucose is present, the operon remains inactive even if lactose is abundant (catabolite repression / positive regulation).

8. Human Genome Project & DNA Fingerprinting

8. मानव जीनोम परियोजना एवं डीएनए अंगुलिछापी

8.1 Human Genome Project (HGP, 1990–2003)

8.1 मानव जीनोम परियोजना

HGP was a 13-year international megaproject coordinated by the U.S. Department of Energy and the National Institutes of Health (NIH), joined by Wellcome Trust (U.K.) and partners in Japan, France, Germany, and China.

Two Methodological Approaches:

1. **Expressed Sequence Tags (ESTs):** Targeted sequencing focused exclusively on identifying all genes that are transcribed into functional RNA.
2. **Sequence Annotation:** The blind, unbiased sequencing of the *entire genome* (both coding exons and non-coding regions), followed by assigning functional boundaries to specific chromosomal regions using computational algorithms.

Total genomic DNA was sheared randomly, cloned into host organisms (bacteria and yeast) using specialized vectors — **BAC (Bacterial Artificial Chromosomes)** and **YAC (Yeast Artificial Chromosomes)** — and sequenced on automated machines using **Frederick Sanger's dideoxy chain-termination method**.

8.2 Salient Features of the Human Genome

8.2 मानव जीनोम के प्रमुख लक्षण (एनसीईआरटी तथ्य)

- The human genome contains $3164.7\ \text{million}$ **base pairs** ($\approx 3.16 \times 10^9\ \text{bp}$).
- Total number of genes is estimated at $\approx 30,000$ (vastly lower than earlier estimates of $80,000$–$140,000$).
- The average gene size is $3,000\ \text{bases}$; the largest known human gene is **Dystrophin** spanning $2.4\ \text{million bases}$.
- Almost all (99.9%) nucleotide bases are exactly identical among all human individuals; individuality stems from merely 0.1% variation!
- Functions are unknown for over 50% of discovered genes.
- **Less than** 2% of the genome codes for functional proteins.
- Repetitive sequences make up a very large portion; they have no direct coding function but illuminate chromosome structure, dynamics, and evolution.
- **Chromosome 1** has the most genes ($2,968$); the **Y chromosome** has the fewest (231).
- Scientists identified about $1.4\ \text{million locations}$ where single-base differences occur: **SNPs (Single Nucleotide Polymorphisms)**.

BOSONANCE

8.3 DNA Fingerprinting

8.3 डीएनए अंगुलिछापी (डीएनए फिंगरप्रिंटिंग)

Pioneered by **Alec Jeffreys**. It relies on identifying variations in non-coding **repetitive DNA**, which is separated from bulk genomic DNA as distinct satellite peaks during density gradient centrifugation:

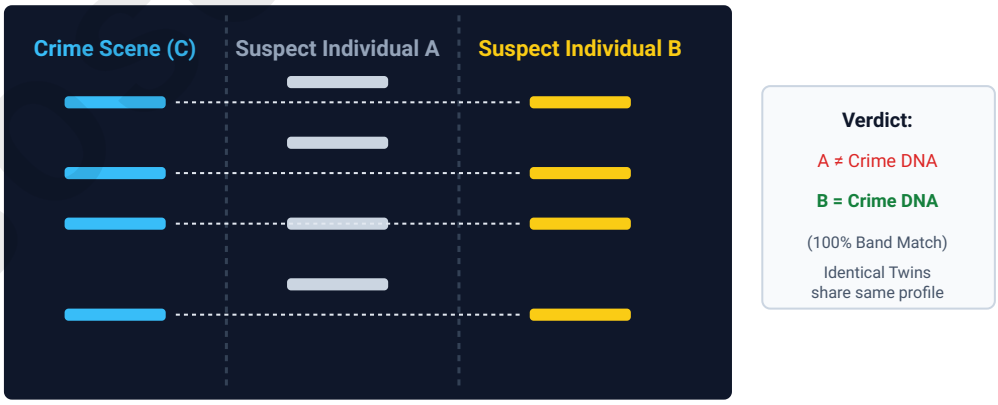
Key Fundamentals of DNA Profiling:

- **Repetitive DNA vs. Satellite DNA:** Bulk genomic DNA forms the major density peak; smaller satellite peaks constitute satellite DNA, categorized into *micro-satellites* and *mini-satellites* based on base composition ($\mathrm{A} : \mathrm{T}$ or $\mathrm{G} : \mathrm{C}$ richness), segment length, and repetition number.
- **Variable Number of Tandem Repeats (VNTR):** Belongs to mini-satellites. A short DNA sequence tandemly repeated many times; the copy number varies widely among chromosomes and individuals, yielding fragment sizes from 0.1 **to** $20\ \mathrm{kb}$.
- **Inheritable DNA Polymorphism:** Mutations accumulating in non-coding sequences without affecting reproductive fitness are inherited stably from parents to offspring, providing the scientific basis for forensic identification and paternity disputes.

Technical Workflow of Southern Blot Hybridization:

1. Isolation and extraction of cellular DNA.
2. Digestion of DNA into fragments by *restriction endonucleases*.
3. Separation of DNA fragments based on molecular size via *agarose gel electrophoresis*.
4. Denaturation and transfer (blotting) of separated DNA fragments onto synthetic membranes (*nitrocellulose or nylon*).
5. Hybridization using **radiolabeled VNTR probes**.
6. Detection of hybridized band patterns via **autoradiography** on X-ray film.

Forensic Autoradiogram Pattern Matching (VNTR Gel Profiling)



Autoradiogram band-matching showing exact concordance between crime scene DNA evidence (C) and Suspect B.

BOSONANCE

9. Standard Worked Quantitative & Analytical Examples

9. मानक हल किए गए संख्यात्मक एवं विश्लेषणात्मक उदाहरण

Worked Example 1 (Chargaff's Rule Calculation):
Problem: A double-stranded DNA sample is chemically analyzed and found to contain 28% Cytosine. Calculate the percentage of Adenine, Guanine, and Thymine in this DNA.
Detailed Solution:

1. According to Chargaff's rules for double-stranded DNA:
$$[\mathrm{G}] = [\mathrm{C}] \quad \text{and} \quad [\mathrm{A}] = [\mathrm{T}]$$
2. Given $[\mathrm{C}] = 28\%$, it directly follows that $[\mathrm{G}] = 28\%$.
3. Total pyrimidine and purine content equals 100%:
$$[\mathrm{G}] + [\mathrm{C}] = 28\% + 28\% = 56\%$$
4. The remaining percentage represents Adenine plus Thymine:
$$[\mathrm{A}] + [\mathrm{T}] = 100\% - 56\% = 44\%$$
5. Since $[\mathrm{A}] = [\mathrm{T}]$:
$$[\mathrm{A}] = \frac{44\%}{2} = 22\% \quad \text{and} \quad [\mathrm{T}] = 22\%$$
6. **Final Answer:** Cytosine $= 28\%$, Guanine $= 28\%$, Adenine $= 22\%$, Thymine $= 22\%$.

Worked Example 2 (DNA Length and Nucleosome Number):
Problem: A mammalian diploid nucleus contains 6.6×10^9 bp. Calculate: (a) the total physical length of the double helix in metres, and (b) the theoretical number of nucleosome core particles present in this cell.
Detailed Solution:

1. **Part (a): Total Physical Length:** The distance between consecutive base pairs in standard B-DNA is $0.34\ \text{nm} = 0.34 \times 10^{-9}\ \text{m}$.
$$\text{Length} = \text{Total base pairs} \times \text{Distance between bp}$$
$$\text{Length} = 6.6 \times 10^9\ \text{bp} \times 0.34 \times 10^{-9}\ \text{m/bp} = 2.244\ \text{metres} \approx 2.2\ \text{m}$$
2. **Part (b): Theoretical Number of Nucleosomes:** One standard nucleosome contains ≈ 200 bp of DNA wrapped around the histone octamer core.
$$\text{Number of nucleosomes} = \frac{\text{Total bp in diploid genome}}{200\ \text{bp/nucleosome}}$$
$$\text{Number of nucleosomes} = \frac{6.6 \times 10^9}{200} = 3.3 \times 10^7\ \text{nucleosomes (33 million beads)}$$

BOSONANCE

Worked Example 3 (Meselson-Stahl Generational Isotope Distribution):
Problem: E. coli cells fully adapted to $^{15}\mathrm{N}$ medium are transferred to $^{14}\mathrm{N}$ medium and allowed to divide for $80\ \mathrm{minutes}$. What will be the proportion of heavy ($^{15}\mathrm{N}/^{15}\mathrm{N}$), hybrid ($^{15}\mathrm{N}/^{14}\mathrm{N}$), and light ($^{14}\mathrm{N}/^{14}\mathrm{N}$) DNA molecules?
Detailed Solution:

1. *E. coli* generation time is $20\ \mathrm{minutes}$. In $80\ \mathrm{minutes}$, the number of generations $n = \frac{80}{20} = 4$.
2. Total DNA molecules produced from 1 initial parental duplex after 4 generations $= 2^4 = 16$ duplexes.
3. Because replication is semiconservative, the 2 original heavy ($^{15}\mathrm{N}$) strands remain conserved as templates forever, yielding exactly 2 **hybrid duplexes ($^{15}\mathrm{N}/^{14}\mathrm{N}$)**.
4. The remaining $16 - 2 = 14$ duplexes consist exclusively of newly synthesized light nitrogen: 14 **light duplexes ($^{14}\mathrm{N}/^{14}\mathrm{N}$)**.
5. Heavy molecules ($^{15}\mathrm{N}/^{15}\mathrm{N}$): 0.
6. **Ratios and Proportions:**
$$\mathrm{Hybrid} : \mathrm{Light} = 2 : 14 = 1 : 7$$
Proportion of Hybrid $= \frac{2}{16} = 12.5\%$; Proportion of Light $= \frac{14}{16} = 87.5\%$.

10. NEET UG Exam-Aligned Practice Assessment

10. नीट यूजी परीक्षा-उन्मुख अभ्यास प्रश्न एवं विस्तृत हल

The following original practice questions adhere strictly to the NEET UG examination standard, incorporating statement-based, assertion-reason, and calculation-oriented patterns.

BOSONANCE

Practice Question 1

English: In a double-stranded DNA segment having 1200 total base pairs, the number of Adenine bases is found to be 360. What will be the total number of hydrogen bonds stabilizing this DNA segment?

A) 2880

B) 3240

C) 3120

D) 3600

हिंदी: एक द्विकुंडलीत डीएनए खंड में कुल 1200 क्षार युग्म (base pairs) हैं, जिनमें एडेनिन क्षारों की संख्या 360 है। इस डीएनए खंड को स्थिर रखने वाले कुल हाइड्रोजन बंधों की संख्या क्या होगी?

A) 2880

B) 3240

C) 3120

D) 3600

Correct Answer: B) 3240

English Solution:

1. Total base pairs $= 1200$, which means there are 1200 paired bases on each strand (total 2400 nucleotides).
2. Number of Adenine (A) bases $= 360$. By Chargaff's rule, Thymine $(\mathrm{T}) = 360$.
3. Total $\mathrm{A} = \mathrm{T}$ base pairs $= 360$. Each $\mathrm{A} = \mathrm{T}$ pair has 2 hydrogen bonds:
$$\text{H-bonds}_{(A=T)} = 360 \times 2 = 720$$
4. Remaining base pairs are $\mathrm{G} \equiv \mathrm{C}$:
$$\text{Number of } \mathrm{G} \equiv \mathrm{C} \text{ pairs} = 1200 - 360 = 840$$
5. Each $\mathrm{G} \equiv \mathrm{C}$ pair is stabilized by 3 hydrogen bonds:
$$\text{H-bonds}_{(G \equiv C)} = 840 \times 3 = 2520$$
6. Total hydrogen bonds $= 720 + 2520 = 3240$.

हिंदी व्याख्या:

कुल क्षार युग्म $= 1200$ हैं। एडेनिन $(\mathrm{A}) = 360$ है, अतः थायमिन $(\mathrm{T}) = 360$ होगा।
A और T के बीच 2 हाइड्रोजन बंध होते हैं: $360 \times 2 = 720$।
शेष $\mathrm{G} \equiv \mathrm{C}$ युग्मों की संख्या $= 1200 - 360 = 840$ होगी।
प्रत्येक G और C के बीच 3 हाइड्रोजन बंध होते हैं: $840 \times 3 = 2520$।
कुल हाइड्रोजन बंध $= 720 + 2520 = 3240$। अतः विकल्प B सही है।

BOSONANCE

Practice Question 2

English: If the sequence of the coding strand of a transcription unit is written as $5'$-A T G C A T G C A T G C\text{-}$3'$, what will be the corresponding sequence of the transcribed mRNA?

A) $5'$-A U G C A U G C A U G C\text{-}$3'$

B) $3'$-U A C G U A C G U A C G\text{-}$5'$

C) $5'$-U A C G U A C G U A C G\text{-}$3'$

D) $3'$-A U G C A U G C A U G C\text{-}$5'$

हिंदी: यदि किसी अनुलेखन एकक के कोडिंग रज्जुक का अनुक्रम $5'$-A T G C A T G C A T G C\text{-}$3'$ है, तो अनुलेखित mRNA का संगत अनुक्रम क्या होगा?

A) $5'$-A U G C A U G C A U G C\text{-}$3'$

B) $3'$-U A C G U A C G U A C G\text{-}$5'$

C) $5'$-U A C G U A C G U A C G\text{-}$3'$

D) $3'$-A U G C A U G C A U G C\text{-}$5'$

Correct Answer: A) $5'$-A U G C A U G C A U G C\text{-}$3'$

English Solution:
By standard molecular convention, the sequence of mRNA is completely identical to that of the **coding strand** ($5' \rightarrow 3'$), with the singular difference that Uracil (U) replaces Thymine (T).
Given Coding Strand: $5'$-A T G C A T G C A T G C\text{-}$3'$.
Replacing T with U yields: $5'$-A U G C A U G C A U G C\text{-}$3'$.

हिंदी व्याख्या:
अनुलेखन के नियमानुसार, बनने वाले mRNA का अनुक्रम कोडिंग रज्जुक ($5' \rightarrow 3'$) के बिल्कुल समान होता है, केवल थायमिन (T) के स्थान पर यूरेसिल (U) आ जाता है। अतः सही उत्तर $5'$-A U G C A U G C A U G C\text{-}$3'$ है।

BOSONANCE

Practice Question 3

English: Consider the following four statements regarding eukaryotic transcription and RNA processing:
I. RNA polymerase I is responsible for transcribing $28S$, $18S$, and $5.8S$ rRNAs.
II. RNA polymerase III transcribes $tRNA$, $5S$ rRNA, and snRNAs.
III. In capping, methyl guanosine triphosphate is attached to the $3'$-end of hnRNA.
IV. Splicing represents the removal of exons and joining of introns in a defined order.
Which of the statements given above are **CORRECT**?

A) I and II only

B) I, II, and III only

C) II and IV only

D) I, III, and IV only

हिंदी: यूकैरियोटिक अनुलेखन एवं आरएनए प्रसंस्करण के संदर्भ में निम्नलिखित चार कथनों पर विचार कीजिए:
I. आरएनए पॉलीमरेज I, $28S$, $18S$ तथा $5.8S$ rRNAs का अनुलेखन करता है।
II. आरएनए पॉलीमरेज III, $tRNA$, $5S$ rRNA तथा snRNAs का अनुलेखन करता है।
III. आच्छादन (कैपिंग) में मेथिल ग्वानोसिन ट्राइफॉस्फेट hnRNA के $3'$-छोर पर जुड़ता है।
IV. समबंधन (स्प्लिसिंग) में व्यक्तेक (एग्जॉन) को हटाया जाता है तथा अव्यक्तेक (इंट्रॉन) को जोड़ा जाता है।
उपरोक्त में से कौन से कथन **सही** हैं?

A) केवल I तथा II

B) केवल I, II तथा III

C) केवल II तथा IV

D) केवल I, III तथा IV

Correct Answer: A) I and II only
English Solution:
- Statement I is CORRECT: RNA Pol I transcribes $28S$, $18S$, and $5.8S$ rRNAs.
- Statement II is CORRECT: RNA Pol III transcribes $tRNA$, $5S$ rRNA, and snRNAs.
- Statement III is INCORRECT: Capping adds methyl guanosine triphosphate to the $5'$**-end** of hnRNA, not the $3'$-end (tailing of 200–300 adenylate residues occurs at the $3'$-end).
- Statement IV is INCORRECT: In splicing, **introns (non-coding)** are removed and **exons (coding)** are ligated together.

हिंदी व्याख्या:
कथन I एवं II पूर्णतः सत्य हैं। कथन III गलत है क्योंकि कैपिंग $5'$-छोर पर होती है, $3'$-छोर पर नहीं। कथन IV गलत है क्योंकि स्प्लिसिंग में इंट्रॉन (अव्यक्तेक) को निकाला जाता है और एग्जॉन (व्यक्तेक) को परस्पर जोड़ा जाता है। अतः विकल्प A सही है।

BOSONANCE

Practice Question 4

English: In a mutation study on the lac operon of *E. coli*, a mutation in the regulatory i gene results in the synthesis of a repressor protein that can permanently bind the operator, but has completely lost its binding site for the inducer lactose. What will be the operational state of the lac operon when lactose is introduced into the growth medium?

A) Operon will remain constitutively active producing all three enzymes.

B) Operon will be permanently repressed; no enzymes will be synthesized.

C) Only β-galactosidase will be synthesized, but not permease.

D) Operon will be transiently induced for only 20 minutes.

हिंदी: *ई. कोलाई* के लैक ऑपेरॉन में एक उत्परिवर्तन अध्ययन के दौरान, नियामक i जीन में ऐसा उत्परिवर्तन होता है जिसके फलस्वरूप उत्पन्न दमनकारी (रिप्रेशर) प्रोटीन प्रचालक (ऑपरेटर) से स्थायी रूप से जुड़ सकता है, किंतु उसने प्रेरक लैक्टोज के लिए अपनी बंधन क्षमता पूर्णतः खो दी है। जब संवर्धन माध्यम में लैक्टोज डाला जाता है, तो लैक ऑपेरॉन की कार्यात्मक स्थिति क्या होगी?

A) ऑपेरॉन निरंतर सक्रिय रहेगा और तीनों एंजाइमों का उत्पादन करेगा।

B) ऑपेरॉन स्थायी रूप से दमित रहेगा; किसी भी एंजाइम का संश्लेषण नहीं होगा।

C) केवल β-गैलेक्टोसिडेज का संश्लेषण होगा, किंतु पर्मिएज का नहीं।

D) ऑपेरॉन केवल 20 मिनट के लिए अस्थायी रूप से प्रेरित होगा।

Correct Answer: B) Operon will be permanently repressed; no enzymes will be synthesized.

English Solution:

Under normal conditions, lactose (allolactose) binds the repressor protein, altering its allosteric conformation so that it detaches from the operator, allowing RNA polymerase to transcribe the structural genes. If the repressor has lost its inducer-binding site, lactose cannot inactivate it. The mutant repressor remains permanently bound to the operator, blocking RNA polymerase continuously. Thus, the operon is **uninducible (permanently repressed)**.

हिंदी व्याख्या:

सामान्य स्थिति में प्रेरक लैक्टोज दमनकारी से जुड़कर उसे निष्क्रिय कर देता है, जिससे वह ऑपरेटर से अलग हो जाता है। यदि दमनकारी में लैक्टोज के लिए बंधन स्थल नष्ट हो चुका है, तो लैक्टोज उसे निष्क्रिय नहीं कर पाएगा। दमनकारी निरंतर ऑपरेटर से जुड़ा रहेगा और आरएनए पॉलीमरेज को आगे नहीं बढ़ने देगा। परिणामस्वरूप ऑपेरॉन स्थायी रूप से दमित (स्विच ऑफ) रहेगा।

BOSONANCE

Practice Question 5

English: Given below are two statements: an Assertion (A) and a Reason (R):
Assertion (A): DNA is chemically less reactive and structurally more stable than RNA, making it a superior genetic material for information storage.
Reason (R): The presence of a $2'$-hydroxyl ($-\mathrm{OH}$) group in every nucleotide of RNA makes it labile and catalytic, whereas the presence of thymine (5-methyl uracil) in DNA provides additional chemical stability.
In light of the above statements, choose the most appropriate answer:

A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

B) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).

C) (A) is true, but (R) is false.

D) (A) is false, but (R) is true.

हिंदी: नीचे दो कथन दिए गए हैं: एक अभिकथन (Assertion A) और दूसरा कारण (Reason R):
अभिकथन (A): डीएनए रासायनिक रूप से कम सक्रिय तथा संरचनात्मक रूप से आरएनए की तुलना में अधिक स्थिर होता है, जो इसे सूचना के भंडारण हेतु श्रेष्ठ आनुवंशिक पदार्थ बनाता है।
कारण (R): आरएनए के प्रत्येक न्यूक्लियोटाइड में $2'$-हाइड्रॉक्सिल ($-\mathrm{OH}$) समूह की उपस्थिति इसे अस्थिर एवं उत्प्रेरकीय बनाती है, जबकि डीएनए में थायमिन (5-मेथिल यूरेसिल) की उपस्थिति इसे अतिरिक्त रासायनिक स्थिरता प्रदान करती है।
उपरोक्त कथनों के संदर्भ में सर्वाधिक उपयुक्त उत्तर का चयन कीजिए:

A) (A) और (R) दोनों सत्य हैं, और (R), (A) की सही व्याख्या है।

B) (A) और (R) दोनों सत्य हैं, लेकिन (R), (A) की सही व्याख्या नहीं है।

C) (A) सत्य है, लेकिन (R) असत्य है।

D) (A) असत्य है, लेकिन (R) सत्य है।

Correct Answer: A) Both (A) and (R) are true, and (R) is the correct explanation of (A).
English Solution:
Both statements are verbatim principles from NCERT Chapter 5. RNA has a reactive $2'$-OH group on ribose making it labile, easily hydrolyzable, and catalytic (e.g., ribozymes). DNA lacks this $2'$-OH group and possesses 5-methyl uracil (thymine), which resists spontaneous deamination and photochemical degradation. Thus, (R) directly explains the structural and chemical stability asserted in (A).

हिंदी व्याख्या:
दोनों कथन एनसीईआरटी के अनुसार पूर्णतः सत्य हैं। आरएनए में $2'$ कार्बन पर उपस्थित $-\mathrm{OH}$ समूह एक क्रियाशील समूह है जो आरएनए को अस्थिर और आसानी से अपघटित होने वाला बनाता है। डीएनए में इस समूह का अभाव तथा यूरेसिल के स्थान पर थायमिन की उपस्थिति डीएनए को अतिरिक्त स्थिरता प्रदान करती है। अतः कारण (R), अभिकथन (A) की सही व्याख्या करता है।

BOSONANCE

Practice Question 6

English: Match the biological scientists in **Column I** with their pioneering milestone discovery in **Column II**:

Column I (Scientist)	Column II (Discovery / Contribution)
p. Alec Jeffreys	1. Semiconservative replication proof in *Vicia faba*
q. Frederick Sanger	2. Developed DNA fingerprinting using VNTRs
r. Herbert Taylor	3. Template-independent enzymatic synthesis of RNA
s. Severo Ochoa	4. Dideoxy automated method for sequencing DNA/proteins

A) p-2, q-4, r-1, s-3

B) p-2, q-1, r-4, s-3

C) p-3, q-4, r-1, s-2

D) p-4, q-2, r-3, s-1

हिंदी: स्तंभ I में दिए गए वैज्ञानिकों का **स्तंभ II** में दी गई उनकी महत्वपूर्ण खोजों से सही मिलान कीजिए:

स्तंभ I (वैज्ञानिक)	स्तंभ II (खोज / योगदान)
p. एलेक जेफ्रीज	1. *विसिया फाबा* में अर्ध-संरक्षी प्रतिकृतियन का प्रमाण
q. फ्रेडरिक सेंगर	2. VNTR आधारित डीएनए अंगुलिछापी का विकास
r. हर्बर्ट टेलर	3. आरएनए का टेम्पलेट-स्वतंत्र एंजाइमी संश्लेषण
s. सेवेरो ओचोआ	4. डीएनए एवं प्रोटीन अनुक्रमण की डाई-डीऑक्सी विधि

A) p-2, q-4, r-1, s-3

B) p-2, q-1, r-4, s-3

C) p-3, q-4, r-1, s-2

D) p-4, q-2, r-3, s-1

Correct Answer: A) p-2, q-4, r-1, s-3

English Solution:

- Alec Jeffreys developed the DNA fingerprinting technique utilizing Variable Number of Tandem Repeats (VNTRs) $\rightarrow$ (2).
- Frederick Sanger developed the chain-termination method for determining nucleotide sequences in DNA and amino acid sequences in proteins $\rightarrow$ (4).
- Herbert Taylor proved semiconservative chromosome replication in *Vicia faba* using radioactive thymidine $\rightarrow$ (1).
- Severo Ochoa discovered polynucleotide phosphorylase facilitating template-independent RNA polymerization $\rightarrow$ (3).

हिंदी व्याख्या:

एलेक जेफ्रीज ने डीएनए फिंगरप्रिंटिंग तकनीक विकसित की (p-2); फ्रेडरिक सेंगर ने डीएनए एवं प्रोटीन अनुक्रमण विधि खोजी (q-4); हर्बर्ट

BOSONANCE

टेलर ने विसिया फाबा पर रेडियोएक्टिव थायमिडीन द्वारा प्रयोग किया (r-1); सेवेरो ओचोआ एंजाइम ने बिना टेम्पलेट आरएनए बहुलकीकरण संभव बनाया (s-3)। अतः सही विकल्प A है।

Practice Question 7

English: Which of the following sets of codons function exclusively as stop (nonsense/terminator) codons during translation, for which no corresponding tRNAs exist?

A) $\mathrm{AUG}, \mathrm{UAA}, \mathrm{UGA}$

B) $\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$

C) $\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGG}$

D) $\mathrm{UGA}, \mathrm{UAG}, \mathrm{UAC}$

हिंदी: निम्नलिखित में से प्रकूटों (कोडॉन) का कौन सा समूह रूपांतरण (प्रोटीन संश्लेषण) के दौरान विशेष रूप से समापक (स्टॉप) प्रकूट के रूप में कार्य करता है, जिनके लिए कोई संगत tRNA उपस्थित नहीं होता?

A) $\mathrm{AUG}, \mathrm{UAA}, \mathrm{UGA}$

B) $\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$

C) $\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGG}$

D) $\mathrm{UGA}, \mathrm{UAG}, \mathrm{UAC}$

Correct Answer: B) $\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$

English Solution:

Out of the 64 genetic codons, 61 code for amino acids. Three codons — UAA **(ochre),** UAG **(amber), and** UGA **(opal)** — do not code for any amino acid and are recognized by Release Factors rather than tRNAs, signaling termination of translation. UGG codes specifically for Tryptophan; AUG codes for Methionine.

हिंदी व्याख्या:

कुल 64 प्रकूटों में से 3 प्रकूट — $\mathrm{UAA}, \mathrm{UAG}, \mathrm{UGA}$ — किसी भी अमीनो अम्ल का कूटलेखन नहीं करते। इन्हें समापक प्रकूट (Stop codons) कहा जाता है। इनके लिए कोई tRNA नहीं होता, बल्कि रिलीज फैक्टर जुड़कर पॉलीपेप्टाइड शृंखला को मुक्त करता है।

BOSONANCE

Practice Question 8

English: In the Human Genome Project, the sequence of which human chromosome was completed last, in May 2006?

A) Chromosome Y

B) Chromosome X

C) Chromosome 1

D) Chromosome 22

हिंदी: मानव जीनोम परियोजना (HGP) में, किस मानव गुणसूत्र का अनुक्रमण सबसे अंत में मई 2006 में पूरा किया गया था?

A) गुणसूत्र Y

B) गुणसूत्र X

C) गुणसूत्र 1

D) गुणसूत्र 22

Correct Answer: C) Chromosome 1
English Solution:
According to NCERT Chapter 5, the sequence of **Chromosome 1** (the largest human chromosome containing $2,968$ genes) was completed only in **May 2006**. This was the last of the 24 human chromosomes (22 autosomes and X and Y) to be fully sequenced.

हिंदी व्याख्या:
एनसीईआरटी के अनुसार, गुणसूत्र 1 (जिसमें सर्वाधिक 2968 जीन हैं) का अनुक्रमण मई 2006 में पूरा हुआ था। यह मानव के सभी 24 गुणसूत्रों (22 अलिंगसूत्र तथा X एवं Y) में से सबसे अंत में अनुक्रमित होने वाला गुणसूत्र था।

11. High-Yield Revision Checklist & NEET Traps

11. महत्वपूर्ण पुनरावलोकन सूची एवं सामान्य भ्रांतियां

Topic / Area	Frequent NEET Student Confusion	Scientific Reality (NCERT Standard)
$^{15}\mathrm{N}$ **Isotope in Meselson-Stahl**	Assuming $^{15}\mathrm{N}$ is a radioactive isotope.	$^{15}\mathrm{N}$ is a **heavy stable isotope**, NOT radioactive! It is separated strictly on density via CsCl centrifugation. (Radioactive thymidine was used by Taylor).
Coding vs. Template Strand	Assuming the coding strand is transcribed into RNA.	The **template strand** ($3' \rightarrow 5'$) is transcribed. The coding strand ($5' \rightarrow 3'$) does not code, but the RNA sequence matches it identical in polarity and sequence (with U in place of T).
The i Gene of Lac Operon	Assuming i stands for *inducer*.	The letter i derives from **inhibitor** (it codes for the repressor protein). Lactose or allolactose is the inducer!
Chargaff's Rule Applicability	Applying $A+G=T+C$ to any nucleic acid.	Chargaff's equivalence holds **only for double-stranded DNA**. It fails for single-stranded DNA ($\phi \times 174$) and all single-stranded RNAs.

BOSONANCE

Topic / Area	Frequent NEET Student Confusion	Scientific Reality (NCERT Standard)
Prokaryotic Ribozyme	Believing all cellular enzymes are proteins.	In bacterial translation, $23S$ **rRNA** functions as the catalytic ribozyme (peptidyl transferase) that forms peptide bonds.
Polyadenylation (Tailing)	Assuming poly-A tail is added according to a DNA template.	Tailing adds $200–300$ adenylate residues to the $3'$-end in a **template-independent** manner.

Quick NEET Memory Checklist:

- ✓ $\phi \times 174 = 5386$ bases (ssDNA); Bacteriophage $\lambda = 48,502$ bp; *E. coli* $= 4.6 \times 10^6$ bp; Human haploid $= 3.3 \times 10^9$ bp.
- ✓ Helix Pitch $= 3.4$ nm; Distance between bp $= 0.34$ nm; Diameter $= 2.0$ nm.
- ✓ Histone core $=$ Octamer rich in Lysine and Arginine; Nucleosome core DNA ≈ 200 bp.
- ✓ Heterochromatin $=$ Dense, dark, transcriptionally inactive; Euchromatin $=$ Loose, light, transcriptionally active.
- ✓ Polymerization direction for DNA polymerase and RNA polymerase is ALWAYS $5' \rightarrow 3'$.
- ✓ RNA Pol I $\rightarrow 28S, 18S, 5.8S$ rRNA; RNA Pol II $\rightarrow$ hnRNA (mRNA precursor); RNA Pol III $\rightarrow$ tRNA, $5S$ rRNA, snRNA.
- ✓ Degenerate codons $= 61$ coding for 20 amino acids; Stop codons $=$ UAA, UAG, UGA (3 codons).
- ✓ Lac genes: $z \rightarrow \beta$-galactosidase, $y \rightarrow$ permease, $a \rightarrow$ transacetylase.
- ✓ HGP: 3164.7 million bp; $\approx 30,000$ genes; Largest gene $=$ Dystrophin $(2.4$ Mb); Chr 1 has 2968 genes; Chr Y has 231 genes; 1.4 million SNPs.
- ✓ DNA Fingerprinting: Alec Jeffreys; VNTRs $=$ minisatellites $(0.1$ to 20 kb); Southern Blotting $\rightarrow$ Restriction digest, Gel electrophoresis, Blotting to nylon, Radiolabeled VNTR probe hybridization, Autoradiography.

12. Concise Final Recap

12. संक्षिप्त अंतिम सारांश

Deoxyribonucleic acid (DNA) stores the complete blueprint of biological life in an antiparallel, double-stranded, right-handed helical conformation stabilized by complementary hydrogen bonds ($\text{A} = \text{T}$ and $\text{G} \equiv \text{C}$) and base stacking. It replicates semiconservatively via an asymmetric replication fork utilizing leading and lagging strand mechanics. Transcription transcribes information from a designated template strand into RNA, which undergoes capping, splicing, and tailing in eukaryotes. The triplet, degenerate, and universal genetic code is translated by ribosomes and tRNA adapters into proteins. In prokaryotes, operons such as the lac operon coordinate metabolic responses to substrate availability via repressor-operator dynamics. Modern genomic advances, including the Human Genome Project and VNTR-based DNA Fingerprinting, harness nucleotide sequence polymorphisms for evolutionary, clinical, and forensic breakthroughs.

डीएनए (डीऑक्सीराइबोन्यूक्लिक अम्ल) जीवन की समस्त आनुवंशिक सूचनाओं का भंडारण एक प्रतिसमांतर, द्विकुंडली संरचना में करता है जो पूरक हाइड्रोजन बंधों एवं बेस स्टैकिंग द्वारा स्थिर रहती है। इसका प्रतिकृतियन अर्ध-संरक्षी विधि द्वारा प्रतिकृति द्विशाख पर सतत (अग्रगामी) एवं असतत (पश्चगामी) रज्जुक के रूप में होता है। अनुलेखन में केवल टेम्पलेट रज्जुक से आरएनए बनता है, जो यूकैरियोट्स में कैपिंग, स्प्लिसिंग और टेलिंग द्वारा परिपक्व होता है। त्रिक, अपह्रासित एवं सार्वभौमिक आनुवंशिक कूट को राइबोसोम और अनुकूलक टी-आरएनए द्वारा प्रोटीन में अनूदित किया जाता है। प्रोकैरियोट्स में लैक ऑपेरॉन जैसी प्रणालियां दमनकारी-ऑपरेटर अंत:क्रिया द्वारा जीन अभिव्यक्ति का सटीक नियमन करती हैं। मानव जीनोम परियोजना एवं वीएनटीआर आधारित डीएनए फिंगरप्रिंटिंग ने चिकित्सा, विकासीय जैविकी तथा न्यायिक विज्ञान में क्रांतिकारी आयाम स्थापित किए हैं।

BOSONANCE